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Article in *Odonatologica* · June 2018

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TSOI – a new index based on Odonata populations to assess the conservation relevance of watercourses in Tunisia

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Received 26th April 2017; revised and accepted 17th April 2018

Abstract. Global ecological conditions existing in the streams of northern Tunisia are very unequal. In consideration of their appropriate management and restoration requirements, breeding populations of 28 Odonata species were studied over an annual cycle in twenty-eight watercourses (22 permanent, 6 intermittent). In total, 7 363 larvae and 337 exuviae were collected. We developed a new indicator, the Tunisian Stream Odonatological Index (TSOI), to assess conservation aspects of Odonata biodiversity in African Mediterranean streams by analysing dragonfly communities. The TSOI operates at the species level. Taxonomic richness, voltinism, endemism, relative taxonomic distinctiveness, and regional conservation status (IUCN) are elements taken into account to design this index. Accordingly, presence of both Maghreb endemic and semivoltine species (*Calopteryx exul*, *Platynemis subdilatata*, *Aeshna cyanea*, *Boyeria irene*, *Gomphus lucasii*, *Onychogomphus costae*, *O. forcipatus*, and *O. uncatus*) may provide useful information about the refuge quality of each habitat type for species with different biological features, and their interest level from a conservation point of view. The results of a first application of this approach are presented in the paper as well. As far as we know, a biodiversity conservation index based on benthic invertebrates in the African Mediterranean area has never before been proposed nor used.

Further key words. Dragonfly, agricultural environments, forest environments, Maghreb, North Africa, Mediterranean area, biological index

Introduction

The presence of a vigorous and diverse odonate fauna will always be a reliable indicator of the stability, health, and integrity of a wetland ecosystem (CORBET 1999). Like their adults, dragonfly larvae are generalized, obligate predators; a characteristic not frequent in the insect orders. Moreover,

compared with other aquatic insects, the Odonata exhibit a great variety in life-cycle patterns (CORBET 1964; NORLING 1984). Some species have a larval stage lasting for more than one year while others complete more than one generation per year (CORBET et al. 2006). Thus, a diverse odonate fauna allowing semi- or partivoltine species to finish their life cycle is a proof for good and stable biotic and abiotic conditions of an aquatic ecosystem throughout the year.

In the 20th century, there were conflicting views regarding the usefulness of odonates as biological indicators for environmental quality (e.g., CARCHINI & ROTA 1982; ORMEROD et al. 1990; CORBET 1993, 1999; SCHMIDT 1985; CLARK & SAMWAYS 1996). Currently, the potential of Odonata as excellent bioindicators is widely acknowledged (HAWKING & NEW 1999; CHOVANEC 2000; HOFMANN & MASON 2005; SAMWAYS 2008; SIMAIKA & SAMWAYS 2008; ARNALDO et al. 2010). They are used as indicators of freshwater health *sensu* good water quality or absence of pollution (OERTLI 2008), ecological integrity (CHOVANEC & WARINGER 2001; SCHULTZ et al. 2003; CHOVANEC et al. 2004; SMITH et al. 2007; FERRERAS-ROMERO et al. 2009; SIMAIKA & SAMWAYS 2009; YALLES SATHA & SAMRAOUI 2017), habitat recovery (SAMWAYS & TAYLOR 2004), and global climate change (OTT 2008). In recent years, several indices based on Odonata communities from different regions have been developed, e.g., OHI (CHOVANEC & WARINGER 2001) and DAI (CHOVANEC et al. 2015) in Central Europe, DBI (SIMAIKA & SAMWAYS 2009) in South Africa, OIWI (KUTCHER & BRIED 2014) in the eastern USA, ORI (GOLFIERI et al. 2016) in Alpine rivers (Italy), ISOI (FERRERAS-ROMERO 2013), and OCIC (BERQUIER et al. 2016) in the Mediterranean area.

However, different approaches have been taken to use Odonata as indicator organisms. Those include the recording of adults (CHOVANEC & WARINGER 2001; SAMWAYS & GRANT 2006; SIMAIKA & SAMWAYS 2009; KUTCHER & BRIED 2014; CHOVANEC et al. 2015; MONTEIRO-JUNIOR et al. 2015; BERQUIER et al. 2016; MARTÍN & MAYNOU 2016), the observation of adults and collection of exuviae (RAEBEL et al. 2010; GOLFIERI et al. 2016) or the sampling of larvae and exuviae (CARLE 1979; HARDERSEN 2008; FERRERAS-ROMERO 2013) to evaluate the ecological quality of fluvial stretches. Likewise, it is generally accepted that the most immediate danger to the Odonata is the threat of destruction and impairment of their larval habitats (WILDERMUTH

1994; CORBET 1999). Hence, sampling of larvae and exuviae may provide better insights into site-specific effects and might help to identify conditions limiting the survival of larvae (GIUGLIANO et al. 2012). Collection of exuviae and larvae is probably the best method to define the breeding assemblage of a given site in regions where they can be reliably identified (GOLFIERI et al. 2016), such as North America, Europe, and the Maghrebi area in North Africa. The odonate faunas of Morocco, Algeria, and Tunisia have been the focus of extensive surveys and are currently well known (SAMRAOUI & CORBET 2000; BOUDOT et al. 2009; SAMRAOUI et al. 2010; KORBAA et al. 2014; EL HAISSOUFI et al. 2015; HAMZAOUI et al. 2015; SAMRAOUI & ALFARHAN 2015; KHELIFA et al. 2016 ; KHELIFA & MELLAL 2017). Although these studies are based mainly on records of adults, the close resemblance to the fauna of southern Europe and the existence of some genera in the area represented by a single species allows the identification of the larvae of almost all species.

An important element of the biological assessment of freshwater conditions is an assessment of the direct or indirect effects of human activities or disturbances (SÁNCHEZ-MONTOYA et al. 2009). Similar to what is known for other parts of the world, the Odonata biodiversity of Maghrebi watercourses is positively associated with the density of the riverine vegetation, and species richness can be negatively correlated with pollution (BOUCHELOUCHE et al. 2015). Agricultural use dominates in most of the catchment areas analysed in this study.

As in similar studies carried out recently in the Mediterranean area (BERQUIER et al. 2016), the main objectives of the present study include: (i) characterization of the Odonata community of Tunisian watercourses, (ii) evaluation of the potential for these communities to be part of the ecological status evaluation of these environments, and (iii) development and testing of a new biological index adapted to assess the dragonfly richness of Maghrebi watercourses and to supplement other assessment systems that are still poorly developed in this part of Africa. Therefore, the Tunisian Stream Odonatological Index (TSOI), designed from the analysis of dragonfly communities in streams flowing through this African country, is proposed as a new approach for assessing aspects of the ecological integrity of northern Tunisian streams. The results of a first application of this approach are presented as well.

Materials and methods

The study was conducted from June 2005 to May 2006 in 28 stretches of northern Tunisian watercourses (Fig. 1). Twenty-two of them were permanent, while the other six were intermittent (Table 1). These streams were situated along a large altitudinal gradient in agricultural or forest areas and ranged from headwaters to lower reaches of rivers and from a pristine conservation state to a strongly altered state due to human activities. Eight environmental variables were measured, belonging to three sets: topographical features, land use, and physic-chemical water properties (Table 2). The latter comprised water temperature, conductivity, salinity, dissolved oxygen, pH, and turbidity and were measured *in situ* using Crison Oxi 330 and Crison MM 40 instruments (Table 3).

Three of the localities studied (Sejnane, Melah, Abid), all of them situated downstream of reservoir dams, showed apparently anomalous mean values of salinity. In the studied stretch of Sejnane stream, the mean was especially low as a consequence of the sandy nature of the basin. In contrast, the salinity was very high in Melah due to outcrops of saliferous Triassic strata crossed by the river. The records taken in Abid were influenced by the proximity of the sea and the entry of seawater into the riverbed at certain times of the year.

During this study, the larval populations of Odonata were systematically sampled from June 2005 to May 2006 at monthly intervals. A sample consisted of five hand-net sub-samples taken in transect across the water body (mesh size 0.5 mm). Four transects spaced at a distance of 100 m were sampled at each site. Hand-net sampling, limited to 0.75 m in depth, was used, as it is well adapted to old riverbeds with dense aquatic vegetation. Observations of adults were recorded as well, using a hand-net for catching and the field guides by D'AGUILAR et al. (1985) and ASKEW (2004) for identification. Exuviae were collected in the shrub vegetation of the banks of the streams without a standardized time for searching. Larvae and exuviae were preserved in 70 % ethanol and identified using a Nikon SMZ800 binocular microscope with an eyepiece micrometre according to the keys of CARCHINI (1983), HEIDEMANN & SEIDENBUSCH (2002), and ASKEW (2004).

Principal component analysis (PCA) was performed to identify similarities and differences among sites with respect to the environmental variables.

Variables were examined for normality and $\log(x + 1)$ transformed. A PCA was performed on the following eight environmental variables representing several catchment characteristics, physicochemical properties, and estimated disturbance (Tables 2, 3). The significance of PCA axes was established by the Monte Carlo permutation test (999 runs). Analysis was performed using PCORD 6.08 software.

Definition of the Tunisian Stream Odonatological Index (TSOI)

To value the refuge quality of Tunisian fluvial courses according to the dragonfly communities that they support and especially their conservation interest, we regarded it as necessary to take the following points into consideration:

1) In undisturbed streams of Tunisia (e.g., Labgâa, Ennour) drainage areas covered by native perennial Mediterranean forest, locally called Khroumir, the dragonfly assemblages are usually represented by four or more families, each of them only with one or two genera that are represented by one or two species (KORBAA et al. 2014).

2) A significant part of the characteristic communities of natural Tunisian fluvial courses is constituted by several species with long life cycles (semi-



Figure 1. Map of northern Tunisia with 21 watercourses used as study sites. Localities with interesting odonate communities are indicated by green (forest) and blue (agricultural environment) points, while those highly influenced by human activity are shown in red. Localities correspond to numbers in Table 1.

Table 1. Coordinates and characteristics of 28 study sites in northern Tunisia. Flow, p – permanent; i – intermittent. Texture of channel substrate, B – blocks; G – rollers; Gr – gravel; S – sand; L – silt. LU – Land use, environment, F – forest; A – agriculture; Alt – Altitude.

Sampling site	Code	DD Coordinates [Decimal degrees]	Alt [m a.s.l.]	Flow	Substrate	LU	Stream size (width range)
1-Joumine Amont	Jam	36.96262N 9.52446E	90	p	G, Gr	F	large (12–35 m)
2-Joumine Aval	Jav	37.02994N 9.66344E	16	p	Gr, S, L	A	middle (5–10 m)
3-Sejnane	Sej	37.19325N 9.57938E	6	p	G, Gr, S	A	large (12–35 m)
4-Douimis	Dou	37.20086N 9.62405E	7	p	Gr, S, L	A	small (1–2 m)
5-Kloufi	Klo	37.19621N 9.58537E	2	i	L	A	small (1–2 m)
6-Melah	Mel	37.10629N 9.54025E	16	p	B, G, Gr, S	A	large (12–35 m)
7-Ksar Mezouar	Kmz	36.78287N 9.33666E	236	p	G, Gr, S	A	small (1–2 m)
8-Béja	Bej	36.76083N 9.19439E	176	p	G, Gr, S; L	A	middle (5–10 m)
9-Abid	Abi	36.86725N 10.72477E	2	p	Gr, S	A	large (12–35 m)
10-Zouiet Mgaiez	Zmg	36.94437N 10.88759E	16	p	Gr, S	A	middle (5–10 m)
11-Medjerda Aval	Mav	37.00019N 10.05633E	8	p	L	A	large (12–35 m)
12-Chafrou	Cha	37.02180N 9.99875E	6	i	Gr	A	small (1–2 m)
13-Henna	Hen	37.15483N 10.05861E	53	i	Gr, S, L	A	small (1–2 m)
14-Amor Amont	Aam	36.92177N 8.74051E	12	p	G, Gr, S	A	middle (5–10 m)
15-Amor Aval	Aav	36.92844N 8.75577E	3	p	Gr, S	A	middle (5–10 m)
16-Kébir (Tabar.)	Keb	36.91899N 8.75425E	9	p	G, Gr, S	A	large (12–35 m)
17-Labgâa	Lab	36.74951N 8.69700E	563	p	B, G, Gr, S	F	small (1–2 m)
18-El Mrij	Mrj	36.75168N 8.69082E	577	p	Gr, L	F	small (1–2 m)
19-Bransia	Bra	36.78105N 8.75175E	588	p	B, G, Gr	F	small (1–2 m)
20-Ennour	Enn	36.80072N 8.65871E	418	p	B, G, Gr, S	F	small (1–2 m)
21-Ghézala	Ghe	36.64313N 8.69852E	229	p	G, Gr, S	A	large (12–35 m)
22-Ellil Amont	Ell	36.69016N 8.73977E	246	p	B, G, Gr, S	A	middle (5–10 m)
23-Ghrib	Ghr	36.61594N 8.68555E	257	i	Gr, S, L	A	small (1–2 m)
24-Kesseb	Kss	36.62263N 9.00522E	130	p	Gr, S, L	A	middle (5–10 m)
25-Maâden Nefza	Maa	36.97006N 9.08503E	38	p	Gr, S, L	F	middle (5–10 m)
26-Bouterfes	Bou	36.96041N 8.94305E	108	i	G, Gr, S	F	middle (5–10 m)
27-Tamra	Tam	37.02538N 9.12805E	57	i	G, Gr, S	F	small (1–2 m)
28-Ziatine	Zia	37.19808N 9.22550E	7	p	S	F	middle (5–10 m)

Table 2. Description of environmental variables measured at 28 study sites in northern Tunisia. Parameters were grouped into three categories: topographical, land use, and physico-chemical variables.

Variables	Code	Description
Topographical: altitude	Alt	Meters above sea level
Land use: Environmental disturbance	EnvD	1 – Autochthonous forest environments well conserved, arboreal vegetation dominated by <i>Quercus</i> spp. 2 – Stream in forest habitat, but woods altered by human activities or woods result from restocking with <i>Pinus halepensis</i> . 3 – Agricultural environments with scarcely altered watercourses and discontinuous though abundant arboreal or shrubby vegetation along the banks that projects a shadow on the water. 4 – Agricultural landscapes with an average level of alteration, little use of chemical fertilizers, presence of macrophyte vegetation (<i>Typha</i> sp., <i>Phragmites</i> sp., <i>Tamarix</i> sp.) 5 – Strongly weathered agricultural environments, high use of chemical fertilizers and livestock breeding.
Physico-chemical:		
Temperature	WT	Water temperature [°C]
Conductivity	Cd	Electric conductivity [μS/cm]
Salinity	S	[‰]
Dissolved oxygen	DO	[mg/l]
pH	pH	pH values
Turbidity	Tb	Nephelometric Turbidity Units [NTU]

voltine) belonging to the Aeshnidae or Gomphidae (KORBAA et al. 2014; SAMRAOUI & ALFARHAN 2015). Their larvae need permanent surface waters, or – concerning *Boyeria irene* (Fonscolombe, 1838) – at least wet microhabitats even during dry, hot summers. This means that permanent watercourses that are on their way to becoming intermittent watercourses due to various human pressures (e.g., water extraction, global warming) bear more conservation value than characteristically intermittent watercourses.

3) There are three Maghreb endemics living in streams: *Calopteryx exul* Selys, 1853, *Platycnemis subdilatata* Selys, 1849, and *Gomphus lucasii* Selys,

Table 3. Environmental disturbance level (EnvD) and mean values of physic-chemical water parameters at 28 study sites of northern Tunisian watercourses recorded during twelve monthly samplings from June 2005 to May 2006. Temp – temperature; DO – dissolved oxygen; pH – pH-value; Cond – conductivity; S – salinity; Tb – turbidity.

Site	EnvD	Temp [°C]	Cond [μS/cm]	S [‰]	DO [mg/l]	pH	Tb [NTU]
Jam	3	19.53	400.68	0.27	9.06	8.23	997.50
Jav	4	19.26	499.22	0.36	8.85	8.52	632.50
Sej	2	20.16	3127.00	0.44	8.95	8.45	1003.33
Dou	3	19.81	3004.42	2.55	9.02	8.36	1130.41
Klo	5	19.66	1795.41	1.55	8.76	8.62	1127.50
Mel	3	19.90	644.50	2.66	8.98	7.38	1497.50
Kmz	3	20.44	568.08	0.42	8.88	8.10	444.16
Bej	2	20.86	607.01	0.46	8.82	8.20	1082.50
Abi	2	20.84	3511.25	0.93	8.84	8.46	923.33
Zmg	5	19.66	1627.60	1.18	9.01	8.43	1148.33
Mav	5	20.37	2430.14	2.06	8.94	8.42	1790.00
Cha	5	20.41	5315.93	4.52	8.92	8.66	432.00
Hen	5	20.53	2583.75	2.19	8.87	8.45	670.00
Aam	2	19.40	417.03	0.30	9.01	8.30	910.00
Aav	4	19.55	444.73	0.32	9.01	8.60	974.16
Keb	2	19.54	394.38	0.29	8.99	8.24	1028.33
Lab	1	15.62	140.20	0.13	9.68	6.91	90.16
Mrj	1	16.21	150.33	0.14	9.57	6.96	118.16
Bra	1	16.68	156.34	0.15	9.50	7.06	195.29
Enn	1	16.30	231.27	0.19	9.57	7.11	692.12
Ghe	3	20.00	656.01	0.47	8.96	8.16	1435.83
Ell	3	17.76	454.58	0.32	9.27	8.23	1695.83
Ghr	4	20.09	1572.43	1.08	8.92	8.63	773.33
Kss	4	20.05	605.25	0.44	8.96	8.40	1110.00
Maa	2	19.81	374.92	0.27	8.98	8.36	1055.83
Bou	2	16.53	361.35	0.26	9.52	8.29	1099.16
Tam	2	16.88	161.67	0.15	9.45	7.04	646.21
Zia	2	18.63	441.66	0.34	9.15	8.28	1067.92

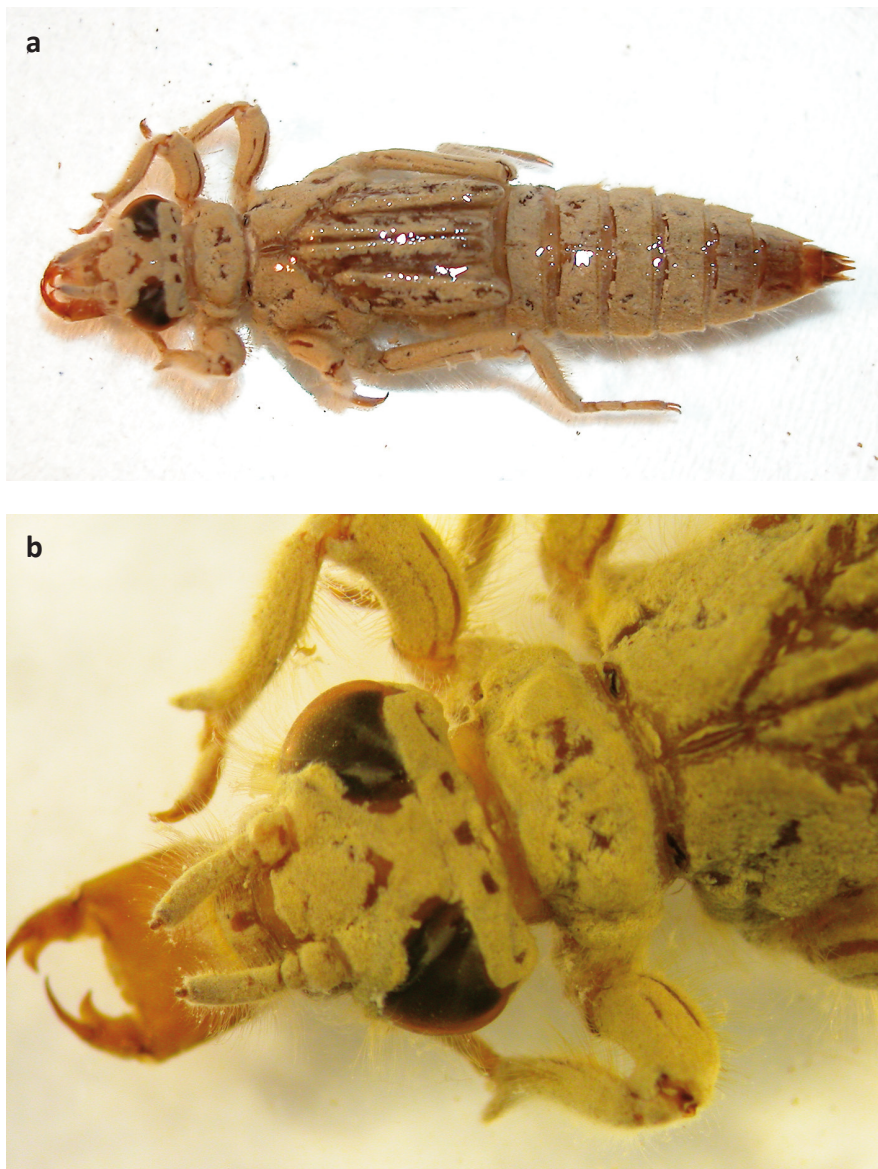


Figure 2. Larva of *Gomphus lucasii*, an endangered and endemic flagship species of natural and near-natural streams in the Maghreb, in dorsal view. a – habitus, b – head (Photos: MK and J. Márquez-Rodríguez).

1849 (Fig. 2). The conservation of sites where these species are present is a priority concern (ZEBBA et al. 2014a; KHELIFA et al. 2015, 2016; KHELIFA 2013, 2016).

The larval stage is probably the most reliable one for obtaining population information (CARLE 1979; HARDERSEN 2008), which is why TSOI calculation is based on species with regularly reproducing populations. Very young larvae cannot be reliably identified to species level. However, odonate communities usually do not comprise many species of the same genus, not even many genera of the same family; this facilitates the faunistic analysis of collected samples. As holds true for the DBI (SIMAIKA & SAMWAYS 2009), the TSOI relies on species presence/absence data. The TSOI comprises two elements: the first is based on the intuitive principle that an assemblage of distantly related species is more diverse than an assemblage of closely related species (WARWICK & CLARK 2001). The second is based on a differentiated valuation of semivoltine species, where each generation requires two years to be completed. Individuals of these species must spend at least one summer in the larval stage. Maghreb endemic species living in streams are also differently rated. Sections of streams where endemics are breeding should be preserved against disturbances by human activities and their conservation must therefore be regarded as a priority concern. Two other elements we considered to positively assess a given watercourse is the harbouring of a dragonfly association including species with high relative taxonomic distinctiveness (RTD) and the presence of threatened species according to the regional conservation status of the IUCN (RLC). For a differentiated valuation of semivoltine and endemic species, the factor 3 is applied to the number of these species found. This factor is the smallest one necessary to achieve a clear discrimination of localities harbouring a significant number of odonate species of conservation interest. This will assist in the preservation of pertinent stream stretches by separating them from stretches that mainly host generalist species without conservation relevance. Furthermore, the application of a factor of 3 allows obtaining a range of values for the TSOI index between 1 and 20, similar to ranges contemplated in other multi-metric indices for the assessment of Mediterranean aquatic fauna communities (e.g., MUNNÉ & PRAT 2009; BERQUIER et al. 2016).

The result of multiplying both elements arrives at a TSOI score per site, calculated by the following equation:

$$\text{TSOI} = (F+G)/2 (3S+3E+R+\Sigma\text{RTD}+\Sigma\text{RLC})/N$$

F – number of families. Seven families were recorded during this study and are used for the calculation of TSOI: Lestidae, Calopterygidae, Platycnemididae, Coenagrionidae, Aeshnidae, Gomphidae, and Libellulidae. Cordulegastridae, as the only other family recently recorded from running waters in Maghreb countries (EL HAISSOUFI et al. 2015), was not included.

G – number of genera. Nine genera of Zygoptera and 10 of Anisoptera (cf. Table 4) are considered for TSOI calculation altogether.

S – number of semivoltine species. Five species reproducing in Tunisian watercourses are regarded as semivoltine: *Aeshna cyanea* (O.F. Müller, 1764), *Boyeria irene*, *Onychogomphus costae* Selys, 1885, *O. forcipatus* (Linnaeus, 1758), and *O. uncatus* (Charpentier, 1840). In North Africa both *A. cyanea* and *B. irene* have obligatory egg diapause in their life cycles (SCHALLER 1960; FERRERAS-ROMERO & PUCHOL-CABALLERO 1995; FERRERAS-ROMERO 1997; GORETTI et al. 2001). *Onychogomphus costae*, *O. forcipatus*, and *O. uncatus* exhibit protracted larval development in similar latitudes of southern Europe (FERRERAS-ROMERO & GARCÍA ROJAS 1995; FERRERAS-ROMERO et al. 1999; CANO-VILLEGAS 2009), which is why they are also considered semivoltine in Maghrebi watercourses. However, it is not clear whether *O. costae* is exclusively or only partially semivoltine in North Africa (ZEBBA et al. 2014b).

E – number of Maghreb endemic species. Three species reproducing in Tunisian streams are endemic (*C. exul*, *P. subdilatata*, *G. lucasii*).

R – number of remaining species.

ΣRTD – relative taxonomic distinctiveness (sum). The relative taxonomic distinctiveness of each species was calculated as follows (FREITAG & VAN JAARSVELD 1997; WHITE et al. 2014): $\text{RTD} = 1 / \sqrt[3]{(\text{family} \times \text{genus} \times \text{species})}$.

ΣRLC – sum of scores assigned here to regionally threatened species (SAMRAOUI et al. 2010). »Endangered« species score 2.0 (*C. exul*, *A. cyanea*), »Vulnerable« species score 1.0 (*Aeshna affinis* Vander Linden, 1823, *G. lucasii*), and »Near threatened« species score 0.5 (*Pyrrhosoma nymphula* (Sulzer, 1776), *B. irene*, *O. costae*).

N – total number of species.

There is only one restriction to the application of TSOI: this index is designed for sites where at least four Odonata species have been collected; a lower number of species could lead to inconsistent rating.

Results

A total of 7363 larvae and 337 exuviae were collected from 28 sampling sites. Twenty-eight species recorded were classified as reproducing, 11 zygopterans and 17 anisopterans, representing 50 % of the specific diversity currently listed for Tunisia (Table 4). The average number of larvae and exuviae recorded at the stations was 275 with a maximum of 1320 (Henna) and a minimum of 30 larvae and exuviae (Ellil Amont). The average reproducing species richness of sampled sites was 9.4 with a maximum of 16 (Ziatine) and a minimum of five species (Joumine Amont, Medjerda Aval, Kesseb, Bouterfes) (Table 5). Three species yielded more than a thousand specimens: *Ischnura graellsii* (Rambur, 1842) (1559), *Platycnemis subdilatata*, (1315), and *Coenagrion caerulescens* (Fonscolombe, 1838) (1202), thus representing more than half of the larvae and exuviae collected (52.9 %). On the other hand, four other species were recorded only as adults: *Coenagrion scitulum* (Rambur, 1842), *Brachythemis impartita* (Karsch, 1890), *Orthetrum cancellatum* (Linnaeus, 1758), and *Trithemis arteriosa* (Burmeister, 1839) (Table 4).

Twenty-two studied sites hosted at least one Maghreb endemic species (Table 5); *P. subdilatata* and *G. lucasii* were found in 21 and 15 sites, respectively. Conversely, *C. exul* was only found in two streams, at elevations between 200 and 250 m a.s.l., flowing through agricultural environments. Twenty sites hosted semivoltine species, but in 12 of those sites *O. forcipatus* was the only representative of that group.

Table 4. List of odonate species at 28 study sites in northern Tunisia where larvae, exuviae or both were collected; (ad) – only adults were collected. Forest sites in bold print. RTD – relative taxonomic distinctiveness of each species; RLC – Threat categories according to IUCN (SAMRAOUI et al. 2010) and scores assigned here to regionally threatened species.

Taxon	Localities	RTD	RLC
<i>Chalcolestes viridis</i> (Vander Linden, 1825)	2, 4, 5, 10, 14, 15, 17, 18, 19, 20, 25, 26, 27, 28	0.29	LC
<i>Lestes virens</i> (Charpentier, 1825)	15	0.14	LC
<i>Calopteryx exul</i> Selys, 1853	7, 22	0.29	EN 2.0
<i>Calopteryx haemorrhoidalis</i> (Vander Linden, 1825)	1 (ad), 8 (ad), 18 (ad), 19 (ad), 27 (ad), 28 (ad)	0.29	LC
<i>Calopteryx</i> sp.	1, 4, 17, 19, 20, 23, 24, 28		
<i>Platycnemis subdilatata</i> Selys, 1849	1, 2, 3, 4, 5, 6, 7, 8, 9, 14, 15, 16, 20, 21, 22, 23, 24, 25, 26, 27, 28	0.35	LC
<i>Ceriagrion tenellum</i> (Villers, 1789)	2, 5, 10, 12, 13, 18, 20, 27, 28	0.19	LC
<i>Coenagrion caerulescens</i> (Fonscolombe, 1838)	2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15, 20, 22 (ad), 23, 28	0.09	LC
<i>Coenagrion scitulum</i> (Rambur, 1842)	13 (ad)		
<i>Erythromma lindenii</i> (Selys, 1840)	2, 3, 4 (ad), 5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 20, 21, 22 (ad), 24, 25, 28	0.13	LC
<i>Erythromma viridulum</i> (Charpentier, 1840)	3, 13 (ad)	0.13	LC
<i>Ischnura graellsii</i> (Rambur, 1842)	2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 21, 22, 23, 24, 25, 26, 27, 28	0.09	LC
<i>Pyrrhosoma nymphula</i> (Sulzer, 1776)	17, 19	0.19	NT 0.5
<i>Aeshna affinis</i> Vander Linden, 1820	13, 17, 19	0.13	VU 1.0
<i>Aeshna cyanea</i> (O.F. Müller, 1764)	13 (ad), 17, 18, 19	0.13	EN 2.0
<i>Anax imperator</i> Leach, 1815	2, 3, 4, 5, 8, 10, 13, 14, 15, 17, 18, 19, 21, 24, 25, 28	0.15	LC
<i>Boyeria Irene</i> (Fonscolombe, 1838)	15, 17, 18, 19, 20	0.26	NT 0.5
<i>Gomphus lucasii</i> Selys, 1849	1, 2, 3, 6, 7, 8, 9, 11, 14, 16, 21, 22, 23, 25, 27 (ad), 28	0.16	VU 1.0
<i>Onychogomphus costae</i> Selys, 1885	6	0.13	NT 0.5

Taxon	Localities	RTD	RLC
<i>Onychogomphus forcipatus</i> (Linnaeus, 1758)	1, 2, 3, 6, 7, 9, 11, 15, 16, 20, 21, 22, 23, 25, 26, 27	0.13	LC
<i>Onychogomphus uncatus</i> (Charpentier, 1840)	14, 16, 17, 18	0.13	LC
<i>Paragomphus genei</i> (Selys, 1841)	1, 3, 6, 14, 15, 16, 25, 27, 28	0.16	LC
<i>Brachythemis impartita</i> (Karsch, 1890)	3 (ad)		
<i>Crocothemis erythraea</i> (Brullé, 1832)	2, 3, 4, 5, 8, 9, 12, 13, 14, 15, 16, 20, 23, 28	0.12	LC
<i>Orthetrum cancellatum</i> (Linnaeus, 1758)	13 (ad)		
<i>Orthetrum chrysostigma</i> (Burmeister, 1839)	3, 4, 5, 7, 9, 10, 12, 13, 14, 15, 16, 22, 23, 25, 26, 27, 28	0.04	LC
<i>Orthetrum coerulescens anceps</i> (Schneider, 1845)	2, 4, 5, 6, 8, 9, 12, 13, 17, 18, 23, 28	0.04	LC
<i>Orthetrum nitidinerve</i> (Selys, 1841)	2, 3, 4, 5, 7, 8, 13, 15, 21, 23	0.04	LC
<i>Orthetrum trinacria</i> (Selys, 1841)	13	0.04	LC
<i>Sympetrum fonscolombii</i> (Selys, 1840)	5, 8, 10, 13, 15, 16, 28	0.05	LC
<i>Sympetrum striolatum</i> (Charpentier, 1840)	2, 5, 10, 12, 13, 23, 28	0.05	LC
<i>Trithemis annulata</i> (Palisot de Beauvois, 1805)	2 (ad), 5, 6 (ad), 11, 12, 13 (ad), 15, 16, 25, 28	0.07	LC
<i>Trithemis arteriosa</i> (Burmeister, 1839)	2 (ad), 3 (ad), 14 (ad)		

The first two axes of the PCA on environmental data where Odonata larvae were found explained 76 % of the variance and distinguished several groups of sampling sites (Fig. 3). Axis 1 explained 65 % and was positively correlated to altitude and the dissolved oxygen concentration and negatively related to environmental disturbance, high mean values of conductivity, pH, and water temperature. Axis 2 explained 11 % and was positively related to salinity and negatively related to turbidity. Axis 1 is the only statistically significant one according to the Monte Carlo test ($p=0.001$). Headwater streams with permanent flow, low mean values of water temperature and conductivity that are within hilly, forest environments (Lab, Mri, Bra, Enn) constitute a first group (Fig. 1). Small streams with intermittent flow, high mean values of salinity, conductivity, and pH (Cha, Hen, Dou) constitute a second group. A third, more complex group, consists of both middle and

Table 5. Characterisation of 28 study sites in northern Tunisia. SR – Species richness, number of species found; Ab – summarised abundance of larvae and exuviae; E – Maghrebi endemic species, S – semivoltine species. For site codes see Table 1.

Site code	SR	Ab	E	S	Site code	SR	Ab	E	S
Jam	5	191	2	1	Aav	15	512	1	2
Jav	13	548	2	1	Keb	11	71	2	2
Sej	12	752	2	1	Lab	10	221	0	3
Dou	10	367	1	0	Mrj	7	112	0	3
Klo	14	275	1	0	Bra	7	105	0	2
Mel	8	130	2	2	Enn	9	119	1	2
Kmz	9	359	3	1	Ghe	7	207	2	1
Bej	10	609	2	0	Ell	6	30	3	1
Abi	9	348	2	1	Ghr	11	275	2	1
Zmg	9	113	0	0	Kss	5	48	1	0
Mav	5	50	1	1	Maa	10	110	2	1
Cha	9	62	0	0	Bou	5	66	1	1
Hen	13	1320	0	0	Tam	7	32	1	1
Aam	11	145	2	1	Zia	16	523	2	0

large streams with permanent flow through agricultural environments (*i.e.*, Bej, Aam, Aav, Keb, Ghe, Ell, Kss) and streams with permanent surface-flow through low altitude, forest environments (Jam, Maa, Bou, Zia).

The resulting TSOI values calculated for the 28 reaches studied ranged between 5 and 17 with a total range amplitude of 12 (Table 6).

Thresholds for five assessment classes are proposed assuming a linear relationship with the odonatological conservation relevance. A scale with the same range amplitude (2.4) was designed to establish the lower thresholds of the five classes (Table 7).

The category ‘high’ was the most frequent class. More than half of the sites (53.6%) had the conservation relevance ‘high’ or ‘good’ (Table 8). On the other hand, only six of 28 studied streams (21.4%) were sites with ‘poor’ or ‘null’ relevance from a conservational viewpoint, with a TSOI score lower than 9.8. Two of them were small, intermittent streams (Chafrou and Henna), and four were permanent streams (Douimis, Zouiet Mgaiez, Medjerda Aval, and Kesseb), all running through agricultural environments.

Sites with a 'good' or 'high' TSOI score above 12.2 within hilly, forest environments over 400 m a.s.l. hosted between five and seven families, two or three semivoltine species, and one or no Maghrebi endemics (Labgâa, El Mrij, Ennour) (Tables 5, 8). In contrast, sites with a TSOI score higher than 12.2, situated within low-altitude forest environments between 5 and 60 m a.s.l., always hosted six or seven families, however, with only one or no semivoltine species and two endemic species (Maâden Nefza, Ziatine). On the other hand, sites with a 'high' TSOI score above 14.6 within agricultural environments harboured five (Ksar Mezouar, Ellil Amont) or six families (Joumine Aval, Amor Amont, Amor Aval), one or two semivoltine

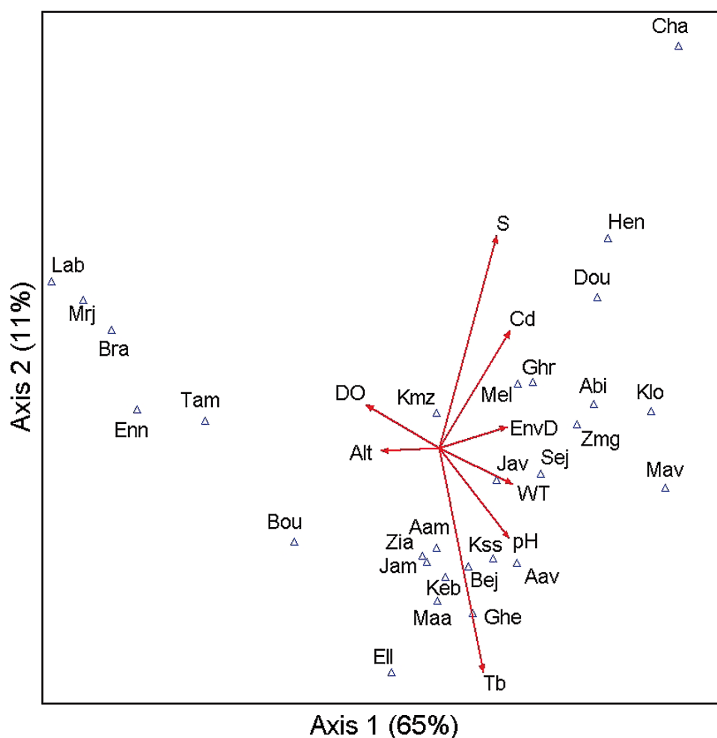


Figure 3. Environmental variability among the northern Tunisian streams, where Odonata were found in this study as described by Principal Component Analysis (PCA). PCA is based on eight environmental variables measured at 28 sites from June 2005 to May 2006. Site and variable codes are as in Tables 1 and 2, respectively.

Table 6. Statistical summary obtained from the values of the TSOI in the eight northern Tunisian watercourses with high biodiversity values: six or seven families are present.

Statistics	TSOI
n	8
Minimum	9.45
Maximum	16.30
Mean	14.68
Standard deviation	2.02
Median	15.28
25 th percentile	14.98
75 th percentile	15.58

Table 7. Thresholds for five classes proposed for the assessment of northern Tunisian watercourses assuming a linear relationship with the odonatological conservation relevance.

Conservation relevance class	TSOI value
High	≥ 14.6
Good	$\geq 12.2 - < 14.6$
Moderate	$\geq 9.8 - < 12.2$
Poor	$\geq 7.4 - < 9.8$
Null	< 7.4

species, and one to three endemic species. Furthermore, sites within agricultural environments and 'good' TSOI scores between 12.2 and 14.6 harboured four or five families, two endemics (*P. subdilatata*, *G. lucasii*), and generally one semivoltine species (*O. forcipatus*).

Odonate species richness in sites considered 'high' or 'good' was very uneven, ranging from six (Ellil Amont) to 16 species (Ziatine); the majority (67%) harboured seven to 11 species. Localities with 'poor' or 'null' TSOI scores below 9.8 hosted two to six families, generally neither semivoltine nor endemic species, and their species richness was very uneven. It ranged from five (Medjerda Aval, Kesseb) to 13 species (Henna), most of which generally were Libellulidae.

Table 8. Number of families, F, and genera, G, relative taxonomic distinctiveness (sum), Σ RTD, regional conservation status of the IUCN (sum), Σ RLC, TSOI scores, and classes obtained at each study site in northern Tunisia.

Sites	F	G	Σ RTD	Σ RLC	TSOI	Class
Joumine Amont	3	5	1.09	1.0	10.47	Moderate
Joumine Aval	6	12	1.83	1.0	15.11	High
Sejnane	5	10	1.59	1.0	12.87	Good
Douimis	6	8	1.50	–	9.45	Poor
Kloufi	5	11	1.70	–	10.11	Moderate
Melah	4	7	1.15	1.5	12.82	Good
Ksar El Mezouar	5	8	1.32	3.0	15.40	High
Béja	5	9	1.22	1.0	11.35	Moderate
Abid	4	8	1.15	1.0	11.43	Moderate
Zouiet Mgaiez	4	8	1.08	–	6.72	Null
Medjerda Aval	3	5	0.58	1.0	8.46	Poor
Chafrou	2	8	0.82	–	5.45	Null
Henna	3	9	1.16	1.0	7.00	Null
Amor Amont	6	11	1.71	1.0	15.29	High
Amor Aval	6	14	2.11	0.5	15.74	High
Kébir (Tabarka)	4	10	1.43	1.0	13.64	Good
Labgâa	6	9	1.74	4.0	16.30	High
El Mrij	5	7	1.19	2.5	14.30	Good
Bransia	4	6	1.44	4.0	11.74	Moderate
Ennour	7	9	1.85	0.5	15.42	High
Ghézala	5	7	1.05	1.0	12.90	Good
Ellil Amont	5	6	1.06	3.0	16.55	High
Ghrib	5	9	1.40	1.0	12.34	Good
Kesseb	4	5	1.01	–	7.21	Null
Maâden Nefza	6	10	1.57	1.0	14.86	High
Bouterfes	5	5	0.90	–	9.90	Moderate
Tamra	5	7	1.25	–	10.50	Moderate
Ziatine	7	14	2.27	1.0	15.27	High

Sites with higher values of relative taxonomic distinctiveness (sum) (Σ RTD) were middle or small streams in both forest (Ziatine, Ennour, Labgâa) and agricultural environments (Amor Aval, Joumine Aval; Table 8), all with

high conservation relevance according to TSOI. Likewise, the streams hosting higher numbers of regionally threatened species populations (Σ RLC) flowed through both forest (Labgâa, Bransia) and scarcely altered agricultural environments, characterized by traditional, extensive agriculture, with little or no use of fertilizers and pesticides, and with moderate management of riparian vegetation (Ksar Mezouar, Ellil Amont; Table 2).

Discussion

The distribution of benthic invertebrate communities depends on multiple environmental factors acting at different spatial-temporal scales (BOYERO 2003). Most biological assessments are based directly or indirectly on the concept of comparing the current condition to hypothetical natural conditions in the absence of human disturbance or alterations (STODDARD et al. 2006). This concept has been adopted, for example, by the European Water Framework Directive (EUROPEAN COMMISSION 2000). The alteration of pristine river conditions at the catchment scale implies changes in habitats and thus in species composition (ALLAN 2004).

The environmental variables affecting Odonata assemblages in lotic habitats are well known (e.g., HARDERSEN 2008; SILVA et al. 2010). Several studies point out that water permanency and flow rate are among the main factors for the composition of Odonata assemblages (CLARK & SAMWAYS 1996; SCHINDLER et al. 2003; BOUDOT et al. 2009). Likewise, disturbance gradients tend to shape the Odonata communities (SEIDU et al. 2017). Thus, it is widely accepted that in the Mediterranean region, altered species assemblages are caused by the fragmentation or loss of riparian forest and canopy, by high water temperatures, by extreme seasonality with drastic flow changes throughout the year, by land-use changes in the surroundings, and by the extraction of groundwater for irrigation (FERRERAS-ROMERO et al. 2009). Together with highly diverse sources of pollution and the consequent absence of pollution-sensitive species, generalist, tolerant species achieve dominance in such disturbed sites (FERRERAS-ROMERO et al. 2009; HAFIANE et al. 2016). Constant water flow, or at least permanent surface waters, are necessary for the survival of long life-cycle species populations, which can also be expected to be sensitive to pollution. An understanding of the relationships between anthropogenic pressures and stream integrity is com-

plex due to the covariation between land-use variables and natural gradients (ALLAN 2004). However, our results discriminate different environmental situations in the northern Tunisian streams we studied. It must be taken into account that water temperature, dissolved oxygen, and even pH values change here throughout the daily cycle. The magnitude of these oscillations will be unequal according to whether they occur in large watercourses or in small streams, or whether the courses are accompanied by riparian forest or devoid of vegetation on the banks. Consequently, the recorded means shown in Table 3 are for guidance only. However, it is also true that in streams that run through forests at altitudes of about 500 m a.s.l. (hilly, forest environments) the recorded means of conductivity, turbidity, water temperature, and pH are the lowest, while the recorded means of DO are the highest. The best variables explaining Tunisian Odonata distribution were: agriculture area vs forested area, water temperature and altitude, intermittent flow vs permanent surface waters, and mineralization and turbidity levels.

This study has yielded a large amount of information on the Odonata communities of Tunisian streams and has improved the overall regional knowledge of these ecosystems. So far, this is the largest data collection campaign completed at the Tunisian national level (AGUESSE 1968; DUMONT 1977; JÖDICKE et al. 2000; KUNZ & KUNZ 2001). Twenty-eight species recorded in this study were considered as reproducing, representing 50% of the specific diversity currently listed for Tunisia; four additional species were recorded only as adults. The average reproducing species richness at sampling sites was higher than that recorded in similar studies previously carried out in this country (SELLAMI et al. 2015). However, species richness was highly variable from one site to another. Among the Zygoptera, *Ischnura graellsii* and *Platycnemis subdilatata* stood out clearly based on their abundances and wide distribution. Likewise, among the Anisoptera, *Anax imperator* Leach, 1815, *Gomphus lucasii*, *Onychogomphus forcipatus*, and *Orthetrum chrysostigma* (Burmeister, 1839) were collected as larva in more than half of locations studied.

Odonatological communities constituted by species with wide distribution on the European continent and generally semivoltine life cycles (*Pyrrosoma nymphula*, *Aeshna cyanea*, *Boyeria irene*, *Onychogomphus uncatus*) that are at the southernmost limit of their range are linked to hilly, forest environments, scarcely mineralized small streams with permanent flow and

well-oxygenated waters. Similar assemblages are known in small streams of forested areas of the southern Iberian Peninsula (FERRERAS-ROMERO & CORBET 1995). In the western Mediterranean area these species are particularly well distributed in altitude (BERQUIER et al. 2016). In contrast, highly mineralized intermittent small streams (Kloufi, Chafrou, Henna, Ghrib) flowing through agricultural environments shelter chiefly populations of libellulids with high ecological tolerances, viz. *Orthetrum coerulescens anceps* (Schneider, 1845), *O. nitidinerve* (Selys, 1841), *O. trinacria* (Selys, 1841), and *Sympetrum striolatum* (Charpentier, 1840).

Gomphidae are considered key components of riverine odonate assemblages (CORBET 1999; FERRERAS-ROMERO et al. 2009; FARKAS et al. 2012; GOLFIERI et al. 2016). Populations of the Maghreb endemic *Gomphus lucasii* and of *Onychogomphus costae*, *O. forcipatus*, and *Paragomphus genei* (Selys, 1841) are associated with large, permanent watercourses running through agricultural land or low altitude forests with high mean values of turbidity. *Onychogomphus forcipatus* was common in streams flowing through forest and through agricultural environments. It is probably the most tolerant semivoltine species in regard to continuity of water flow throughout the year (FERRERAS-ROMERO & MÁRQUEZ-RODRÍGUEZ 2014).

A similarly wide range as that of *G. lucasii* was found for another endemic species, *Platycnemis subdilatata*. Both species exhibit wide tolerance ranges in relation to mineralization and turbidity and had populations in stretches where the annual mean values were 3511 $\mu\text{S}/\text{cm}$ (Abid) for conductivity, 2.66‰ (Melah) for salinity, and approximately 1700 NTU (Ellil Amont, Medjerda Aval) for turbidity. They were even found in two stretches running through strongly weathered agricultural environments. Both endemics can be considered well adapted to the currently impaired conditions of running waters, both in agricultural areas and forest.

However, populations of *Calopteryx exul*, the third Maghreb endemic, were only found in two permanent streams with scarcely altered watercourses (Environmental disturbance level 3). This underlines why this species is currently attributed with the highest level of threat, »Endangered« (RISERVATO et al. 2009).

Some agricultural habitats (Joumine Aval, Kluofi, Amor Aval) had high species richness of odonates. The high abundance of species from the Libel-

lulidae in the agricultural habitat was probably due to the preponderance of anthropogenic disturbances there (SEIDU et al. 2017). This finding agrees with other studies that recorded species richness of odonates to be higher in agricultural habitats (SAMWAYS 1989; FERREIRA-PERUQUETTI & FONSECA-GESSNER 2003) but also that species of less conservation value are dominant. However, mature forest habitats are typically species-poor (KINVIG & SAMWAYS 2000; SEIDU et al. 2017) and proved to be adequate environments for semivoltine anisopterans (El Mrij, Bransia, Labgâa). The dense canopy cover of the forest habitat provides a conducive environment to most forest specialist in this area (*P. nymphula*, *A. cyanea*, *A. affinis*, *B. irene*), which are of higher conservation concern than generalist species. In general, mature forest habitat served as a refuge for most unique and specialized species, while the disturbed habitats were mostly inhabited by eurytopic generalist species (CLAUSNITZER 2003; SEIDU et al. 2017).

In fluvial systems of the Mediterranean area, as in the southern Iberian Peninsula or on the island of Corsica, the mean number of species per locality is usually low (FERRERAS-ROMERO 1999; BERQUIER et al. 2016), but the total diversity of Odonata in those systems is high because the species turnover differ significantly from locality to locality (DE KNIJF & DEMOLDER 2010). The 'Tunisian Stream Odonatological Index' (TSOI) that has been developed in this study is mainly based on (i) the taxonomic diversity (families, genera, and species of Odonata present), (ii) the different forms of voltinism because the existence of semi- or partivoltine populations proves good biotic and abiotic conditions throughout the year, and (iii) biogeographical characteristics, *i.e.*, the presence of endemic species. Relative taxonomic distinctiveness and the presence of threatened species according to the regional conservation status of the IUCN are also considered.

TSOI has a dual objective: to encourage the conservation of endemic species and to contribute to the evaluation of global environmental quality. This new index should be calculated from data pertaining to larvae and exuviae identified to species level. The TSOI score can be obtained from only one sample or more than one. In this study, data were collected by monthly samples throughout the same year, which then were summarized. Possibly, three summarized samples obtained in different seasons within the same hydrologic or natural year, *e.g.*, late autumn, late winter, and late spring or

early summer, should be considered sufficient for an adequate assessment using the TSOI (FERRERAS-ROMERO 2013).

The evaluation carried out according to the TSOI highlights that many of the sampled sites are interesting from a conservation point view regarding Odonata biodiversity. This result could be influenced by factors such as the limited size of the watersheds, the low human population density in major parts of the country, or the relatively low intensity of agricultural land use within most of the territory involved in this study. Locations ranked as having 'null' or 'poor' conservation relevance (Zouiet Mgaiez, Chafour, Henna, and Douimis) are situated close to the sea coast, where human activities are more intense (cattle raising areas and areas of high agricultural production), similar to other areas of the Mediterranean (BERQUIER et al. 2016). The damage to sites of conservation relevance could be primarily related to water extraction for irrigation and waste-water discharge into rivers.

The highest TSOI values were usually found in sites with co-existing populations of semivoltine and Maghreb endemic species (Table 9), harbouring six or seven odonate families, in both forest and agricultural environments. No component of the TSOI is absolutely decisive to define the conservation relevance class. In each specific locality, the numbers of families and genera, of semivoltine species, and of endemics present have important weight for the calculated TSOI score. Thus, sampling site 4 (Douimis) obtains a TSOI score indicating only 'poor' conservation relevance although it harbours six odonate families and eight genera.

The TSOI was specifically developed in the geographical context of North Africa. Application of this index to fluvial systems in the other Maghrebi countries, Morocco and Algeria, would probably only require slight adjustment of the TSOI at most.

The composition of Tunisian dragonfly communities could be useful to detect some types of general environmental alterations. Positive correlation of larval odonate diversity and abundance with overall aquatic macro-invertebrate diversity and abundance has been confirmed in other countries (FOOTE & RICE 2005; HORNUNG 2005; FERRERAS-ROMERO 2013). As far as we know, a biological quality index based on benthic invertebrates in the African Mediterranean area has neither been proposed nor used before. Quantitative assessment of the odonate community breeding in stretches of fluvial cours-

Table 9. Presence of populations of semivoltine species, Maghrebi endemics, and threatened species at relevant sites of this study that could be recommended for conservation management.

Habitat type – TSOI class	<i>A. cyanea</i>	<i>B. irene</i>	<i>O. forcip</i>	<i>O. uncatus</i>	<i>C. exul</i>	<i>P. subdil</i>	<i>G. lucasii</i>	<i>A. affinis</i>
Forest – High								
Labgâa, perm., small, 563 m a.s.l.	+	+		+				+
Ennour, perm., small, 418 m		+	+			+		
Maâden Nefza, perm., middle, 38 m			+			+	+	
Ziatine, perm., middle, 7 m						+	+	
Forest – Good								
El Mrij, perm., small, 577 m	+	+		+				
Agricultural – High								
Joumine Aval, perm., middle, 16 m			+			+	+	
Ksar Mezouar, perm., small, 236 m					+	+	+	
Amor Amont, perm., middle, 12 m				+		+	+	
Amor Aval, perm., middle, 3 m		+	+			+		
Ellil Amont, perm., middle, 246 m			+		+	+	+	

es could add complementary information to assessments obtained through highly accepted biotic indices based on the existence and diversity of sensitive groups of organisms to amounts of dissolved oxygen, water temperature, and degree of eutrophication among other strongly related parameters when these indexes have been adapted to these countries in the future.

Acknowledgements

This work was partially funded by the Agency for International Cooperation, Spanish Ministry of Foreign Affairs and Cooperation, Inter-University Program and Scientific Research Cooperation (PCI-Mediterranean, 2006. B/5823/060). Our thanks to an anonymous reviewer for useful comments and to Florian Weihrauch for improving the text.

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